

DBMS Project 5

USER PROFILE WITH ASSIGNMENT STATISTICS AND DATA
VISUALIZATION

Agenda

1. Introduction
 - 1.1 Requirements
 - 1.2 Database Schema
2. Design
 - 2.1 User Profile
 - 2.2 Task and Assignment Statistics
 - 2.3 Data Visualization
3. Implementation
4. Demonstration

Introduction – Requirements



1. Profile Display

Show username and user role
(Admin or Regular User)



3. Data Visualization

Display stats using charts

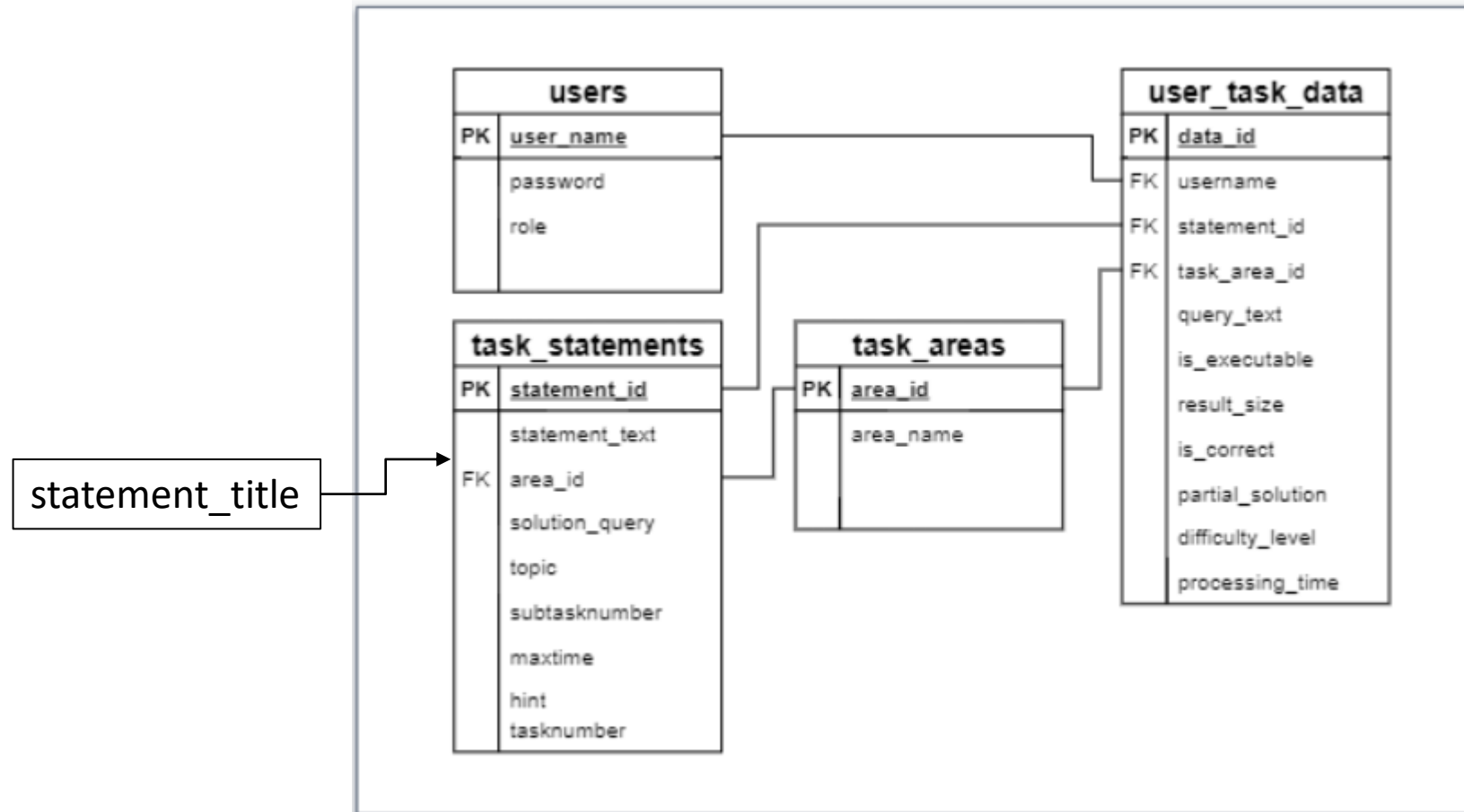


2. Task and Assignment Statistics

- Number of tasks completed per assignment
- Hardest and easiest tasks
- Time spent per task - including longest and shortest
- Optional: Add additional infos and stats

Use PERN Stack

Introduction – DB Design



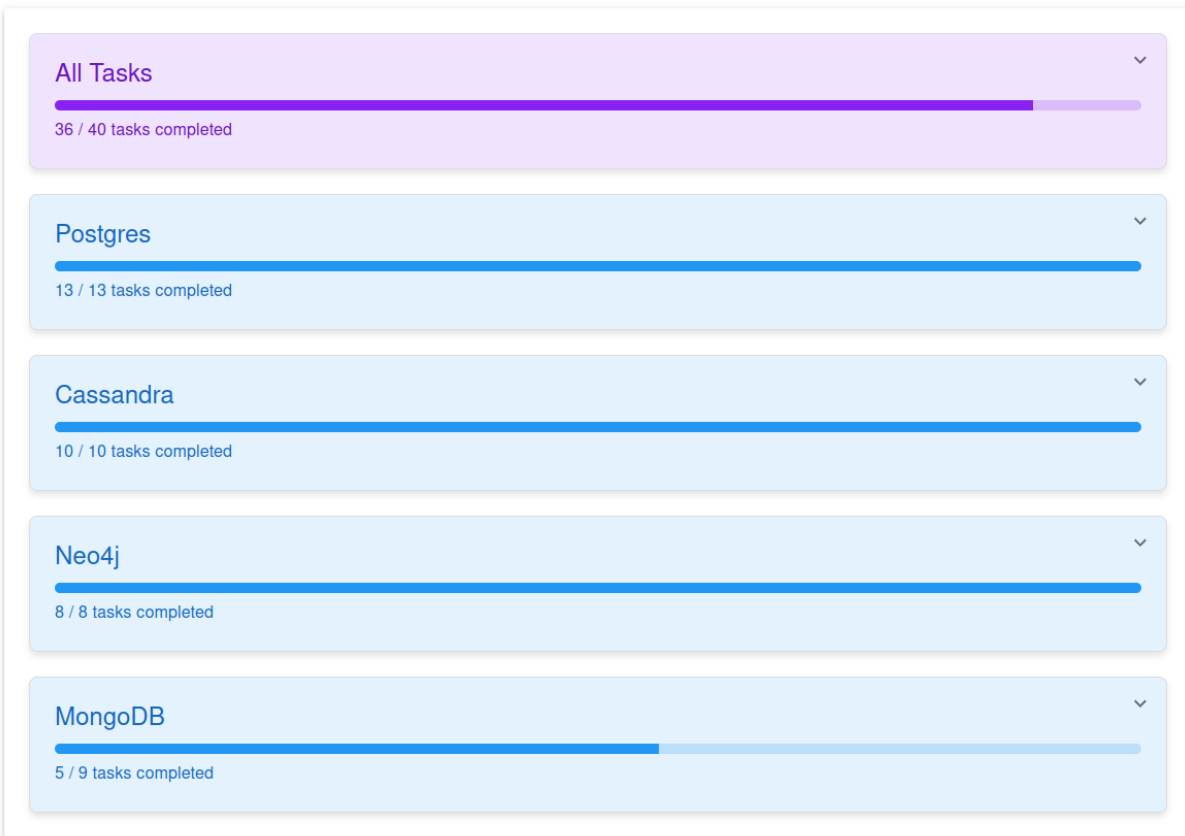
User Profile



bob

Regular User

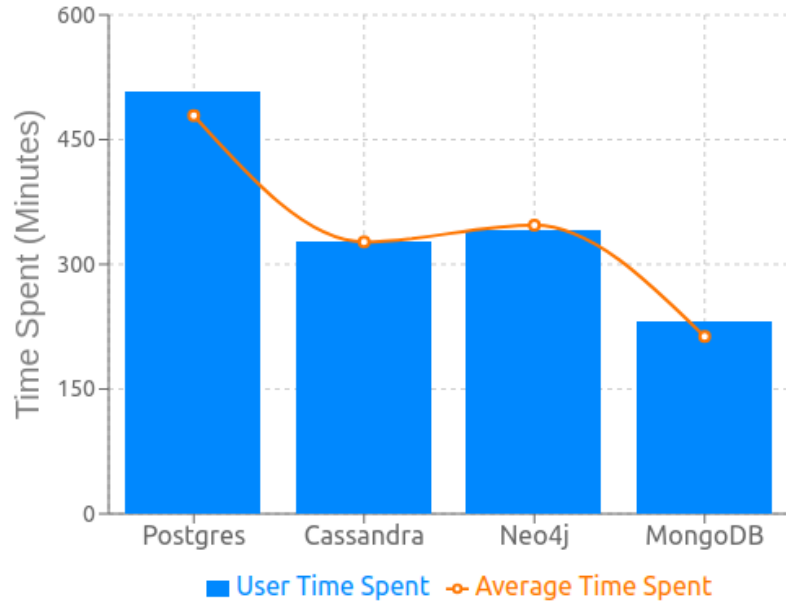
Design – User Profile



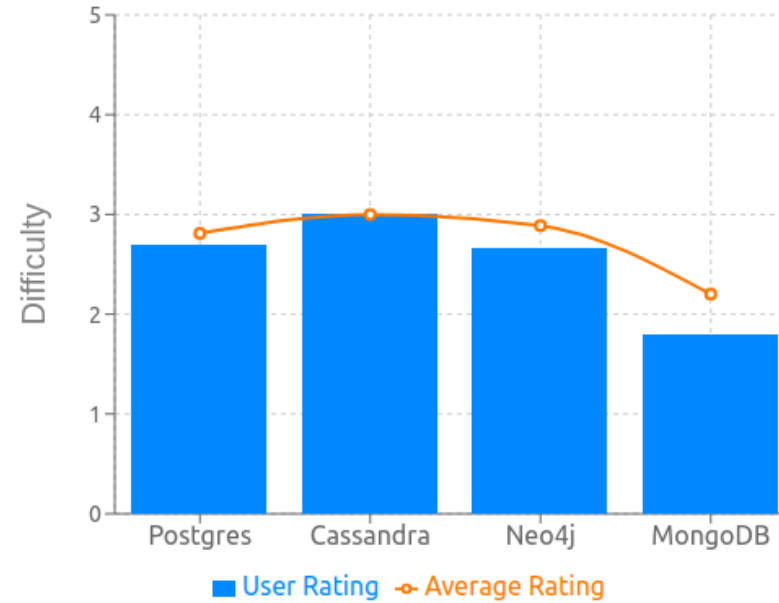
Postgres				
13 / 13 tasks completed				
Task ID	Task	Status	Difficulty ↑	Time Spent
3	Number of emails received per department	✓	Very Easy	25m
5	Update the schema	✓	Very Easy	21m
9	Emails between two values for certain department	✓	Very Easy	25m
6	Add information for entity set with default value	✓	Easy	42m
7	Find missing values	✓	Easy	20m
11	Network size by "knows" relation	✓	Easy	20m
12	Two-hop email network	✓	Easy	42m
1	List of people with their department	✓	Normal	20m
2	Number of emails sent out per department	✓	Normal	15m
8	Emails between two dates	✓	Normal	15m
4	Correlation between salary and number of emails	✓	Very Difficult	1h 50m
10	Network size by e-mail	✓	Very Difficult	1h 50m
13	Social network	✓	Very Difficult	42m

Design – Task and Assignment Statistics

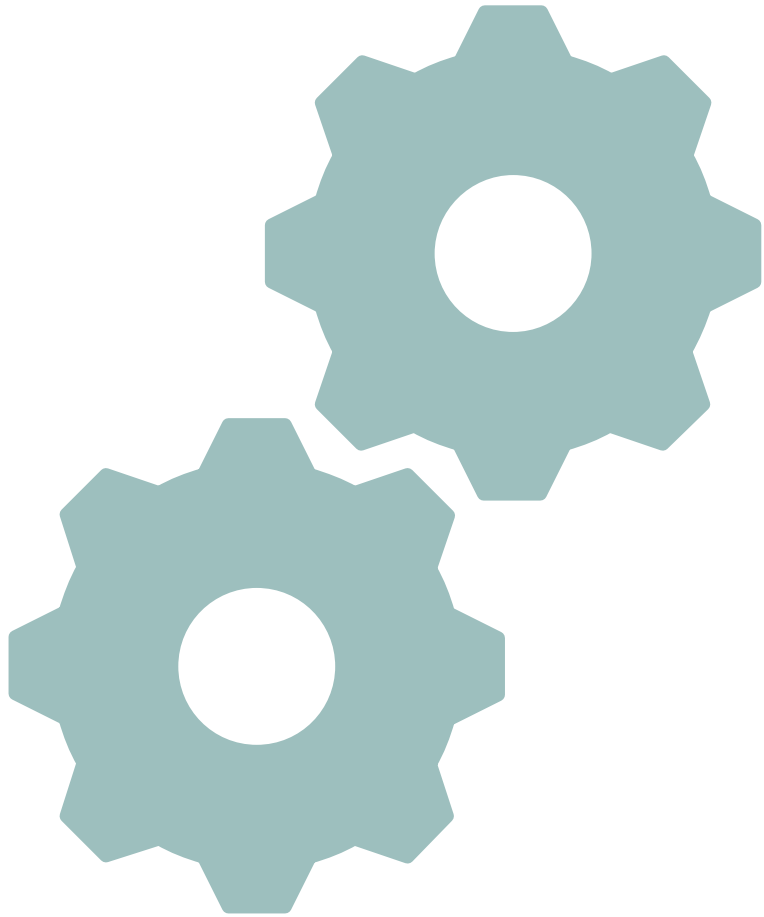
Time Spent on Assignments by User vs Average



Average Difficulty Rating by User vs Average



Design – Data Visualization



Implementation

- PERN + MUI + Recharts
- Three components:
 - UserProfile
 - UserTaskCompletion
 - UserTaskAnalytics
- JS is mostly boiler plate + drawing boxes


```

....WITH
....user_totals AS (
.....SELECT
.....utd.task_area_id,
.....u.user_name,
.....SUM(utd.processing_time) AS user_total_time
.....FROM user_task_data utd
.....JOIN users u ON utd.username = u.user_name
.....WHERE u.role = 'Regular User'
.....GROUP BY utd.task_area_id, u.user_name
....),
....global_averages AS (
.....SELECT
.....task_area_id,
.....AVG(user_total_time) AS global_avg
.....FROM user_totals
.....GROUP BY task_area_id
....),
....personal_totals AS (
.....SELECT
.....ta.area_id,
.....ta.area_name,
.....COALESCE(SUM(utd.processing_time), 0) AS total_time
.....FROM task_areas ta
.....LEFT JOIN user_task_data utd
.....ON ta.area_id = utd.task_area_id
.....AND utd.username = $1
.....GROUP BY ta.area_id, ta.area_name
....)
....SELECT
.....pt.area_id,
.....pt.area_name,
.....pt.total_time,
.....COALESCE(ga.global_avg, 0) AS global_avg
....FROM personal_totals pt
....LEFT JOIN global_averages ga
....ON pt.area_id = ga.task_area_id
....ORDER BY pt.area_id;

```

Implementation - Total Time Spent per Assignment



Demonstration

Thank You for your attention
